

Limitations: proposed changes for review

Toward Personalized Hebrew ASR · Dolev Abudi and Hadas Yonat · 7 October 2026

This note shows every change proposed to the paper’s Limitations section, against the version on `main` (7 October 2026). The rest of the paper was edited in a separate pull request, so this section can be reviewed on its own. Part 1 marks the changes, and part 2 gives the proposed section as it would read.

How to read it: ~~struck-out red text~~ is removed, underlined blue text is added, and black text is unchanged.

What changed, and why

- *The reference is the protocol* and *Reference transcripts* said the same thing (the old TODO asked to keep one). They are merged into one paragraph. It keeps the first one’s text and adds the three points only the second one made: what stenographers do to the speech, that the 0.95 threshold was set with the `ivrit.ai` team, and that comparisons between models scored against the same references are the most reliable.
- *meeting-disjoint* becomes *session-disjoint*, the word used everywhere else in the paper.
- *Limited compute* and *Few speakers beyond 80 minutes* are shorter, with the same content. The future-work clause of *A random control group* is dropped because the Conclusion already lists similar-speaker groups as future work.
- No semicolons.
- Checked: the 75–82% figure is right (74.7% of the clean test and 81.8% of the full test come from committees the same speaker appears in during training). The \$200 total has no source in the repository (the documented costs come to about \$90), so it is worth confirming.

Part 1: the changes, marked

Limitations

Prior exposure. Arm B was fine-tuned on Knesset plenary speech, which includes many of our speakers. Our evaluation uses committee recordings, a different setting and register, but the model may still be partly familiar with these voices.

Shared recording conditions. Train and test sets are ~~meeting-disjoint~~session-disjoint, but not committee-disjoint: about 75–82% of the test speech comes from committees the same speaker also appears in during training, and so likely from the same rooms and microphones. Part of an adapter’s gain may therefore reflect familiar recording conditions rather than the voice alone.

The reference is the protocol. Every WER is measured against the stenographic protocol, which is not a verbatim transcript: stenographers condense, rephrase and omit speech, so part of every measured error is a mismatch between protocol and audio rather than a recognition mistake. Our gains are gains in agreement with the protocol, and may partly reflect learning what the stenographer omits. We limit this by training and validating only on segments whose protocol closely matches the audio (quality ≥ 0.95 , a threshold set with the `ivrit.ai` team, Appendix A). That score, however, comes from a Whisper-family aligner, so a test set filtered the same way would favour speech that Whisper models already transcribe well; and we therefore score every model both on the high-quality segments and on all segments at quality ≥ 0.7 .

Reference transcripts. Our references are the Knesset’s stenographic protocols, not verbatim transcripts: stenographers condense, rephrase and omit speech, so part of every measured error is a mismatch between protocol and audio rather than a recognition mistake. We limit this with alignment quality filters set in consultation with the ivrit.ai team (Section 3.4, Appendix A), but no verbatim No verbatim reference was available. Absolute WERs therefore partly reflect the protocol’s editing, and, so our comparisons are most reliable between models scored against the same references.

Limited compute. ~~Training and inference~~ All experiments ran on paid cloud services (RunPod GPUs and Hugging Face inference), ~~at a total cost of for~~ about \$200 ~~across all our experiments. Within this budget we focused on the experiments central to our question, and a larger budget would allow more~~ in total, which limited the models, configurations and speakers ~~to be explored~~ we could explore.

Tuning on validation loss. The rank, dropout, augmentation and learning rate were chosen on the validation loss of four speakers. The rejected settings, including rank 16, were never scored on the test sets.

Few speakers beyond 80 minutes. The longer budgets cover four speakers, two of them up to 24 hours, and the rank-32 check uses one seed, so the finding that the personal margin does not grow with more speaker data ~~holds for these speakers, and~~ may not hold for every speaker.

A random control group. The control pools speakers at random, so it does not test whether *similar* speakers could replace a speaker’s own audio; ~~we leave that comparison to future work~~.

Part 2: the section as it would read

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